

To compile and run

To Compile:

```
gcc Assignment5.c -o proj2
```

Note: I completed this assignment with gcc version gcc Ubuntu 11.4.0-1ubuntu1~22.04.3) 11.4.0

To Run:

```
./proj2 <inputfile.txt> <scheduling_algorithm> <time_quantum>
```

- Provided input text files:
 - “input.txt” – same input as example in assignment description
 - “input2.txt” – 5 processes, all arrive at the same time
 - “input3.txt” – 3 processes, idle time occurs
 - “input4.txt” – empty file
- Scheduling options:
 - “FCFS”
 - “SJF”
 - “RR”
- Time Quantum
 - Optional if you run FCFS or SJF
 - Required if running RR, provide an int

Test Cases

Test 1

- Provided input from assignment description
- 4 processes
- All arrive in order of pid, 1 time unit between each

FCFS

FCFS					
PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
1	2	12	16	4	10
2	3	16	17	1	13
3	4	17	19	2	13
Average waiting time: 9.00					

SJF

SJF					
PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
2	3	12	13	1	9
3	4	13	15	2	9
1	2	15	19	4	13
Average waiting time: 7.75					

RR, q = 5

Starting RR scheduling			
PID	Start Time	End Time	Running Time
0	0	5	5
1	5	9	4
2	9	10	1
3	10	12	2
0	12	17	5
0	17	19	2

RR					
PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	7	19	12	7
1	2	5	9	4	3
2	3	9	10	1	6
3	4	10	12	2	6

Average waiting time: 5.50

Test 2

- 5 processes
- All arrive at time 0

FCFS

FCFS					
PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
1	0	0	8	8	0
2	0	8	12	4	8
3	0	12	14	2	12
4	0	14	20	6	14
5	0	20	22	2	20
Average waiting time: 10.80					

SJF

SJF					
PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
3	0	0	2	2	0
5	0	2	4	2	2
2	0	4	8	4	4
4	0	8	14	6	8
1	0	14	22	8	14
Average waiting time: 5.60					

RR, q = 3

Starting RR scheduling			
PID	Start Time	End Time	Running Time
1	0	3	3
2	3	6	3
3	6	8	2
4	8	11	3
5	11	13	2
1	13	16	3
2	16	17	1
4	17	20	3
1	20	22	2

RR					
PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
1	0	14	22	8	14
2	0	13	17	4	13
3	0	6	8	2	6
4	0	14	20	6	14
5	0	11	13	2	11
Average waiting time: 11.60					

Test 3

- 3 processes
- Includes idle time

FCFS

Starting FCFS scheduling

Time 0: idle
Time 1: Running task 1
Time 2: Running task 1
Time 3: Running task 1
Task 1 ended, Time: 4
Time 4: idle
Time 5: Running task 2
Time 6: Running task 2
Time 7: Running task 2
Time 8: Running task 2
Time 9: Running task 2
Task 2 ended, Time: 10
Time 10: Running task 3
Time 11: Running task 3
Time 12: Running task 3
Time 13: Running task 3
Task 3 ended, Time: 14

FCFS

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
1	1	1	4	3	0
2	5	5	10	5	0
3	10	10	14	4	0

Average waiting time: 0.00

SJF

Starting SJF scheduling

Time 0: idle
Time 1: Running task 1
Time 2: Running task 1
Time 3: Running task 1
Task 1 ended, Time: 4
Time 4: idle
Time 5: Running task 2
Time 6: Running task 2
Time 7: Running task 2
Time 8: Running task 2
Time 9: Running task 2
Task 2 ended, Time: 10
Time 10: Running task 3
Time 11: Running task 3
Time 12: Running task 3
Time 13: Running task 3
Task 3 ended, Time: 14

```

SJF
PID  Arrival Time  Start Time  End Time  Running Time  Waiting Time
1    1             1           4         3             0
2    5             5           10        5             0
3    10            10          14        4             0
Average waiting time: 0.00

```

RR, q = 2

```

Starting RR scheduling
PID  Start Time  End Time  Running Time
Time 0: idle
1    1          3          2
1    3          4          1
Time 4: idle
2    5          7          2
2    7          9          2
2    9          10         1
3    10         12          2
3    12         14          2

```

```

RR
PID  Arrival Time  Start Time  End Time  Running Time  Waiting Time
1    1             1           4         3             0
2    5             5           10        5             0
3    10            10          14        4             0
Average waiting time: 0.00

```

Test 4

- Empty file

```

kmfirth23@Kennedy-Laptop:~/CS4420/CS-4420-Assignment5$ ./proj2 input4.txt F
CFS
Invalid file format
Nothing to process

```